

Two wires in quadrature

Hybrid A- Φ finite-element solver. Two parallel copper wires driven with equal currents **a quarter cycle apart** — the only case in the suite whose field is **elliptically polarized**, and the one that demonstrates the `[postprocess]` options.

```
regression_tests/06_TwoWires_Quadrature · 1 A per wire at 0° and  
90° · 2.5 kHz · 192,368 tets · 448,438 unknowns · backward  
error 5.3e-22 · 19.8 s
```

0. What this case establishes

Three things, in order of how much work they do.

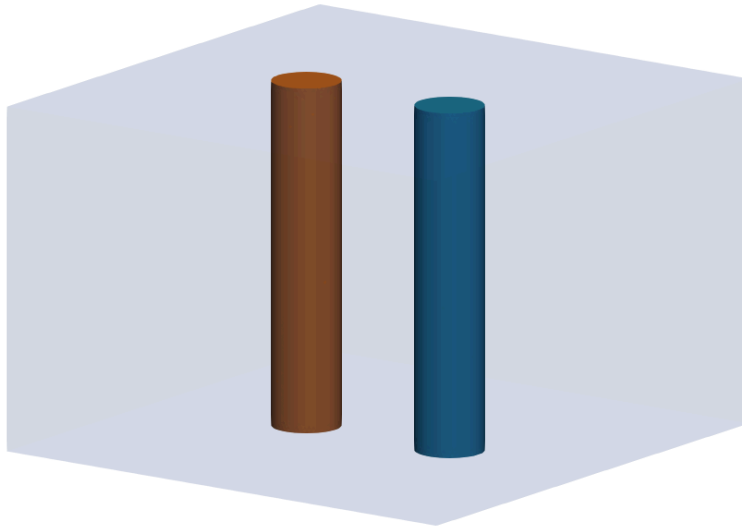
It demonstrates every `[postprocess]` option. Sixteen requests covering all three geometries, every display, `phase_deg`, `colormap` and `data`. That is the reason the case exists, and §4 is the tour.

It is the only case in the suite with an elliptically polarized field. A single conductor has `B` purely azimuthal, so the polarization ellipse collapses to a line and `axial_ratio`, `peak` against `complex_magnitude`, and `phase` all go trivial — case 05's median axial ratio is `9e-06`. Two sources 90° apart make `B` genuinely rotate, and the two-wire superposition says exactly where. §6.

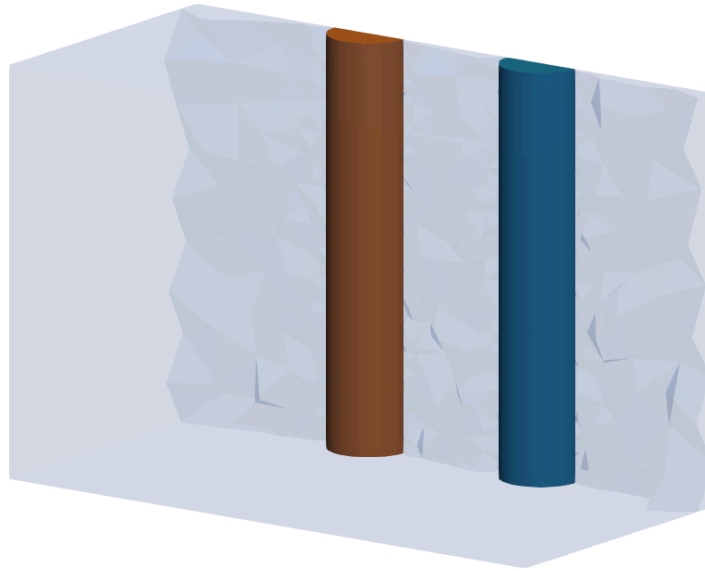
It is the first case to exercise a port `phase_deg` at all. Everything else in the suite uses `0` or omits the key, so nothing would have caught it quietly doing nothing. Measured: **90.014°** between the wires.

It also found a renderer bug on its first run, which is what a case with a known analytic answer is for. `axial_ratio` read `2.7e-05` where theory says exactly 1.0 — `tools/postprocess.py` was dividing `|P×Q|` by `a` once, which gives the semi-minor axis `b` rather than the ratio `b/a`. The wrong quantity is in the field's own units and looks entirely plausible on a colour bar; only a case with an exact expected value separates the two.

1. The geometry



Two copper wires in a box of air. $a = 2 \text{ mm}$, centres $d = 10 \text{ mm}$ apart, 20 mm long, in a $30 \times 30 \times 20 \text{ mm}$ box. The wires run the full height of the box, so their end faces lie *in* the box's end faces — which is what lets each wire be driven by a boundary port rather than needing an internal cut (case 05's problem).



Cut at $y = 0$, the plane through both axes, so the wires are seen directly rather than through 14 mm of air.

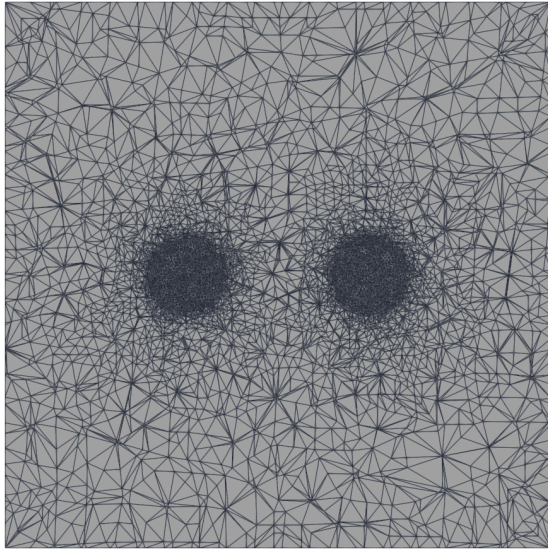
Wire radius a	2 mm
Separation d	10 mm, so $d/a = 5$
Length L_z	20 mm, $L_z/a = 10$
Domain	$30 \times 30 \times 20$ mm of air, $\mathbf{n} \times \mathbf{A} = 0$ on the outer boundary
Conductor	copper, $\sigma = 5.8e7$ S/m
Air	$\sigma = 0$
Frequency	2500 Hz
Skin depth δ	1.3217 mm, so $a/\delta = 1.511$

Why 2.5 kHz. Low enough that the case solves in twenty seconds, high enough that $a/\delta > 1$ and the current visibly crowds toward the surface. It also puts a 53° phase gradient *inside* each conductor, which is what gives $\text{display} = \text{phase}$ something to show in §4.3 instead of a flat fill.

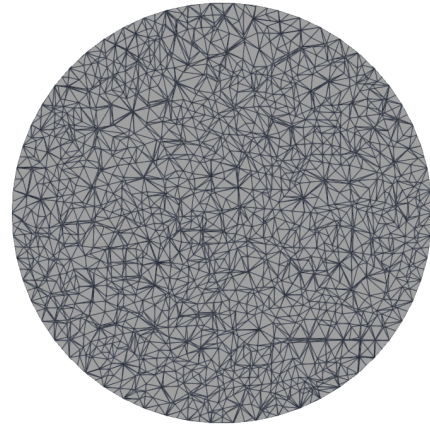
Three bodies, not two. `wire1`, `wire2` and `air` carry `body_tag` 1, 2 and 3. Worth stating because every earlier case has exactly two bodies and the air is always tag 2 — scripts that hard-code that assumption read a wire here and produce a plausible, wrong answer.

`tools/skin_profile.py` finds the conductors by `sigma > 0` instead of by tag for this reason.

2. The mesh, and which one



The `xy` cut at mid-height. Fine inside the conductors, coarsening outward through the air to `lc_far = 3 mm` at the walls.



Wire 1 alone, at the same `z`. Median element **0.317 mm** against `δ = 1.3217 mm` — about **4.2 elements per skin depth**.

The size *field* asks for 0.25 mm and the median comes out 0.317 mm. These are not the same number and the report quotes the measured one. gmsh treats `lc_wire` as a target that its own quality constraints relax, so a resolution claimed from the `.geo` is optimistic by about 25 % here.

2.1 The refinement has to go *inside* the conductor

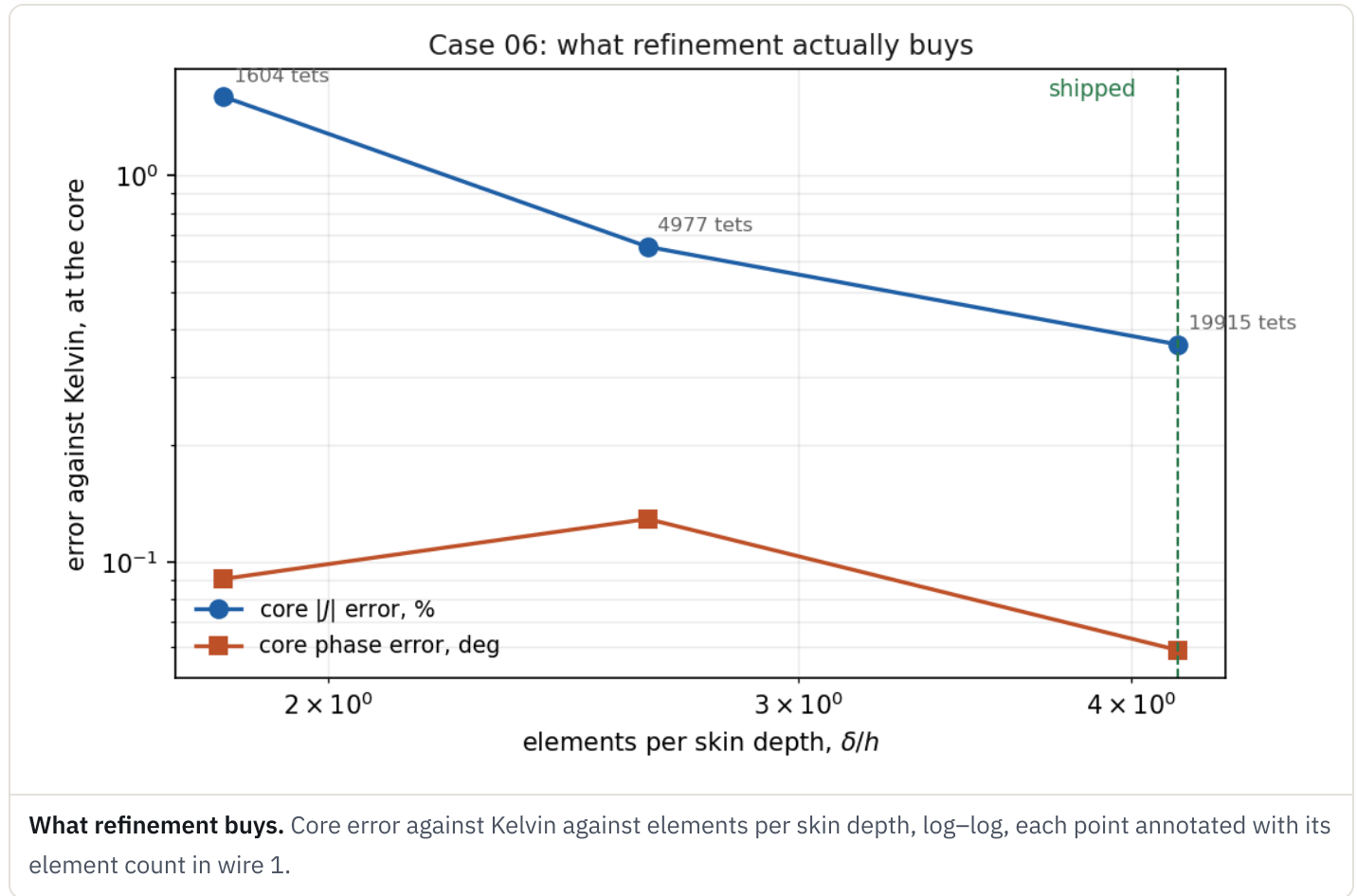
The skin depth is a length scale in the metal, so that is where the elements must be small. A distance-to-the-surface field cannot express that: distance is positive on **both** sides of a surface, so asking for 0.25 mm at the wire boundary also refines a 2 mm shell of **air** around each wire. That produced **about 660,000 elements, more than half of them air** — air in which nothing varies on a skin-depth scale at all.

Two gmsh `Cylinder` fields instead, which size the interior of a cylinder and leave everything outside it alone, with the air graded on its own terms:

```
Field[3] = Cylinder;
Field[3].Radius = 1.02 * a;
Field[3].VIn = lc_wire; // 0.25 mm
Field[3].VOut = lc_far;
```

192,368 tets instead of 660,000, for the same resolution where it matters.

2.2 Which mesh — measured, not asserted



Three refinements, each meshed, solved and compared against Kelvin at the core by

`make_convergence_data.py` :

LC_WIRE	MEDIAN H	PER Δ	WIRE TETS	CORE $ $ ERR	CORE LAG ERR	SOLVE
0.60 mm	0.724 mm	1.8	1,604	+1.600 %	+0.090°	~8 s
0.40 mm	0.502 mm	2.6	4,977	+0.654 %	+0.129°	~12 s
0.25 mm	0.317 mm	4.2	19,915	+0.365 %	-0.059°	~20 s

The shipped mesh is the finest, and the magnitude column is what justifies it: 1.60 % → 0.65 % → 0.37 %, for 8 s → 20 s. The coarsest is already fine for a demonstration you edit and re-run; the finest is what makes the skin-effect comparison in §5 worth quoting at all.

The phase column is not converging, and that is honest rather than worrying. It wanders $+0.090^\circ \rightarrow +0.129^\circ \rightarrow -0.059^\circ$. A tenth of a degree is this comparison's own noise floor — the radial binning, the azimuthal spread within each bin and the finite cell sample each contribute more than the discretisation does. A clean monotone trend at that magnitude would be more suspicious than this wandering is.

Three points do not establish a convergence order, and this report does not claim one.

An earlier version of this case claimed first-order convergence in h from *two* meshes. Two points fit any power law you care to assume — they cannot test one. The third point showed the apparent order collapsing to 0.37, which turned out not to be a property of the solver at all but a constant offset in the measurement: §5.3.

3. The drive

Each wire is driven independently, and each therefore needs its own potential reference: a current source on the bottom face, a 0 V terminal on the top.

```
[port P1]                                [port P3]
type      = boundary_current              type      = boundary_current
surface    = wire1_bottom                 surface    = wire2_bottom
current    = 1.0                          current    = 1.0
phase_deg  = 0                           phase_deg  = 90      # <-- the whole point

[port P2]                                [port P4]
type      = boundary_voltage              type      = boundary_voltage
surface    = wire1_top                   surface    = wire2_top
voltage    = 0.0                         voltage    = 0.0
```

Under the solver's $e^{+j\omega t}$ convention a port phase ϕ means the terminal current is $I \cos(\omega t + \phi)$, so wire 2's current leads wire 1's by a quarter cycle.

	MEASURED	TARGET
I in wire 1	1.000405 A	1 A
I in wire 2	0.998795 A	1 A
phase difference	90.014°	90°

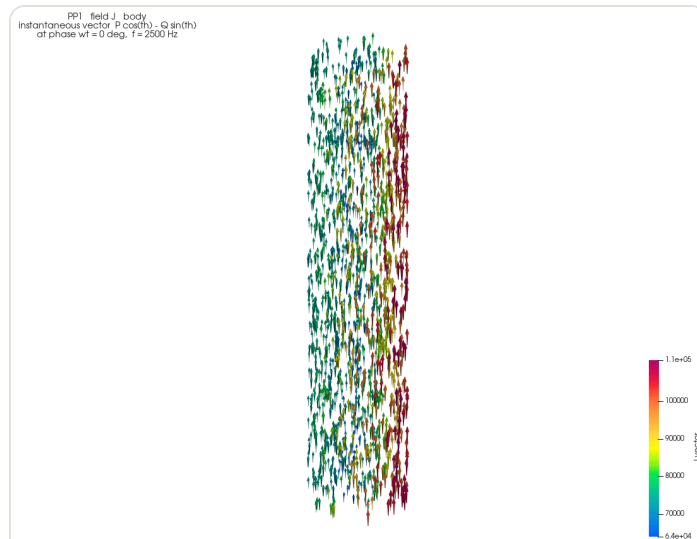
The current is recovered by integrating J_z over a **slab** at mid-length and dividing by the slab's thickness — $\sum J_z \Delta V / \Delta z$ — rather than over a single plane of faces. Many layers of tets then contribute and no one plane's mesh can set the answer. Neither wire's port is on the plane being measured, so this is a check and not a restatement of the input.

4. The post-processing tour

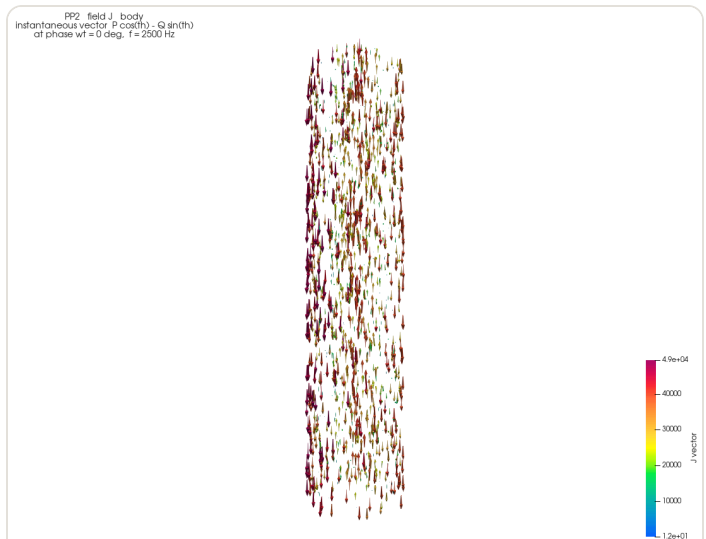
Sixteen requests. Every picture in this section comes from the manifest the solver itself writes (`output/postprocess.json`) rendered by `tools/postprocess.py` . Nothing here is hand-plotted — that is the claim the case is making, so `collect_postprocess_figs.py` copies those exact files into `fig/` rather than redrawing them.

```
..\run_case.bat two_wires.aphi
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" ..\..\tools\postprocess.py output
```

4.1 The current, as arrows



PP1 — **geometry = body**, **field = J**, **display = vector**, **phase_deg = 0**. A side view of the whole conductor; every arrow points along **+z** and is coloured by **|J|**, from **6.4e4** to **1.09e5 A/m²**. At this instant wire 1 is at its peak: every cell is positive and the net is **+0.9995 A**.



PP2 — the same instant, **wire 2**, whose terminal current is at its zero crossing. The net instantaneous current really is zero: **-0.0003 A**. But **J** is not zero anywhere — see below.

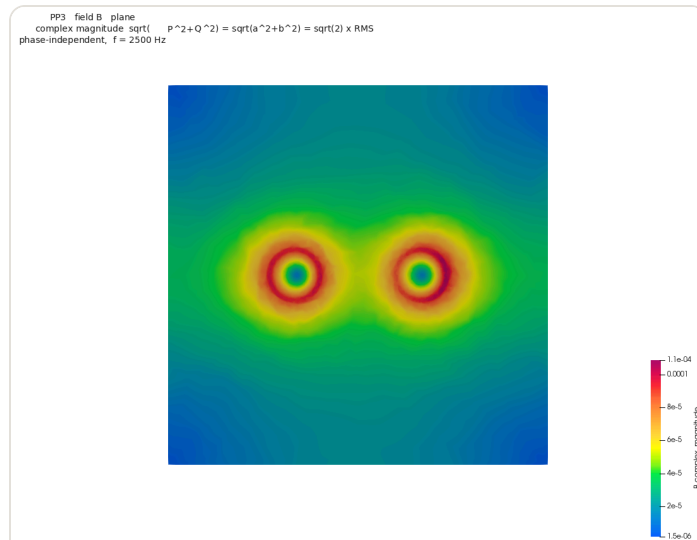
In PP1, read **across** the silhouette, which is a projection through the wire: the left edge is the face turned away from wire 2 and the right edge is the face turned toward it. The gradient between them is the **proximity effect**, quantified in §5.2. The skin effect is in this picture too but is harder to read here, because a projection superimposes the near and far surfaces on the middle of the silhouette — §5.1 is where it is measured properly.

PP2 is the more interesting of the two. Wire 2's net current is zero at this instant, but **J** inside it reaches **$\pm 4.6e4$ A/m²** — 40 % of wire 2's own phasor peak of **1.15e5** — and **the outer shell and the core flow in opposite directions**: measured here, cells with **$J_z < 0$** occupy **$r/a = 0.64 \dots 0.98$** and cells with **$J_z > 0$** occupy **$r/a = 0.01 \dots 0.76$** . The two cancel in the integral and not pointwise.

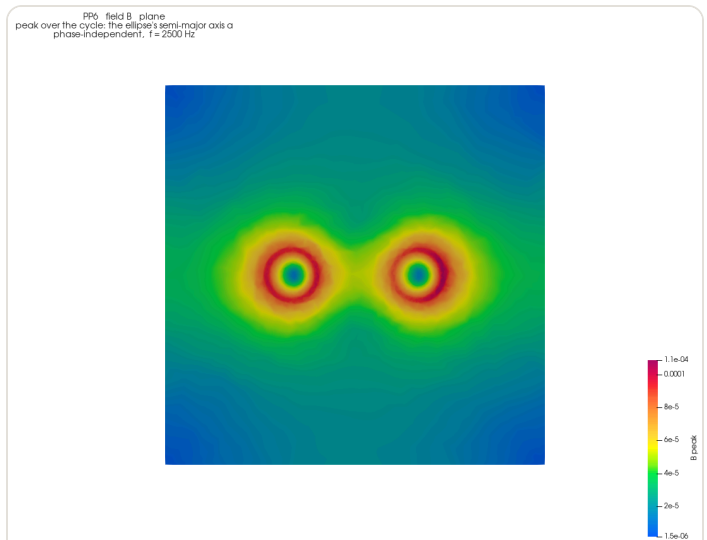
That is the skin effect stated in the time domain. **J** lags by 53° more at the core than at the surface (§5.1), so the radii cannot all cross zero together; at the instant the *terminal* current vanishes, what is left inside is a purely **circulating** eddy current. The arrows read as downward in the picture because the outer shell dominates a projection — it is both the larger magnitude and the longer path through the silhouette.

4.2 The three magnitudes, one field

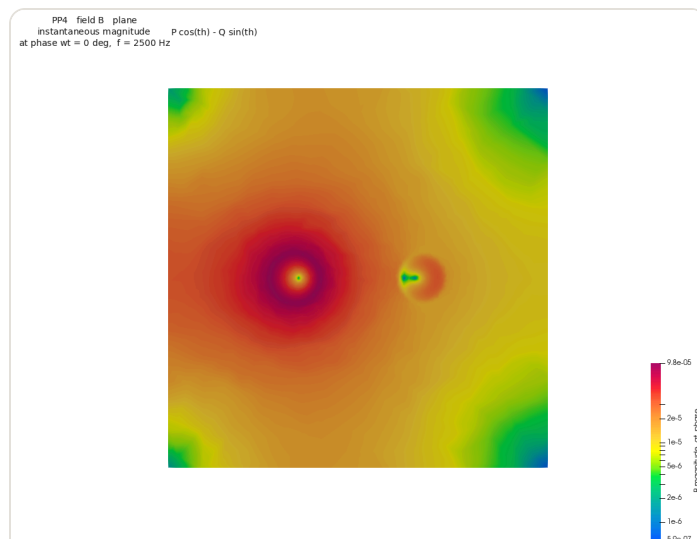
The heart of the tour: three pictures of the same **B** on the same plane, none of them wrong, each answering a different question.



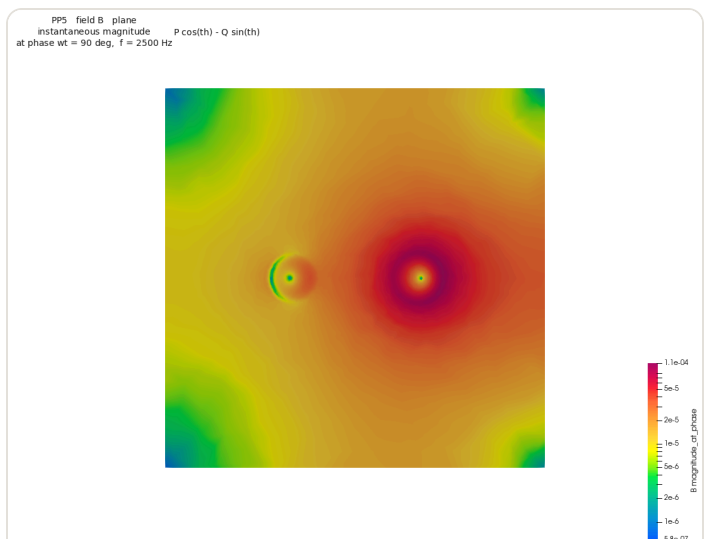
PP3 — **display = complex_magnitude** (the default). **N**
 $= \sqrt{|P|^2 + |Q|^2} = \sqrt{2} \times \text{RMS}$, independent of phase.
Left-right symmetric about the perpendicular bisector,
 because both wires carry the same amplitude and **N**
 does not care that they are 90° apart. **|B|** peaks in the
 red ring at each conductor's *surface* and falls toward its
 axis, as it must. This is the one that never blinks, which is
 why it is the default.



PP6 — **display = peak**, the largest **|B|** ever reaches
 over the cycle: the semi-major axis **a** of the polarization
 ellipse. Compare it with PP3 — where the field is
 circularly polarized the two differ by $\sqrt{2}$, and §6
 measures exactly that.



PP4 — **display = magnitude_at_phase**, **phase_deg = 0**.
 A single instant, $|P \cos \theta - Q \sin \theta|$ at $\theta = 0$.
 Wire 1 is at its peak and wire 2 near zero, so the
 symmetry of PP3 is gone.



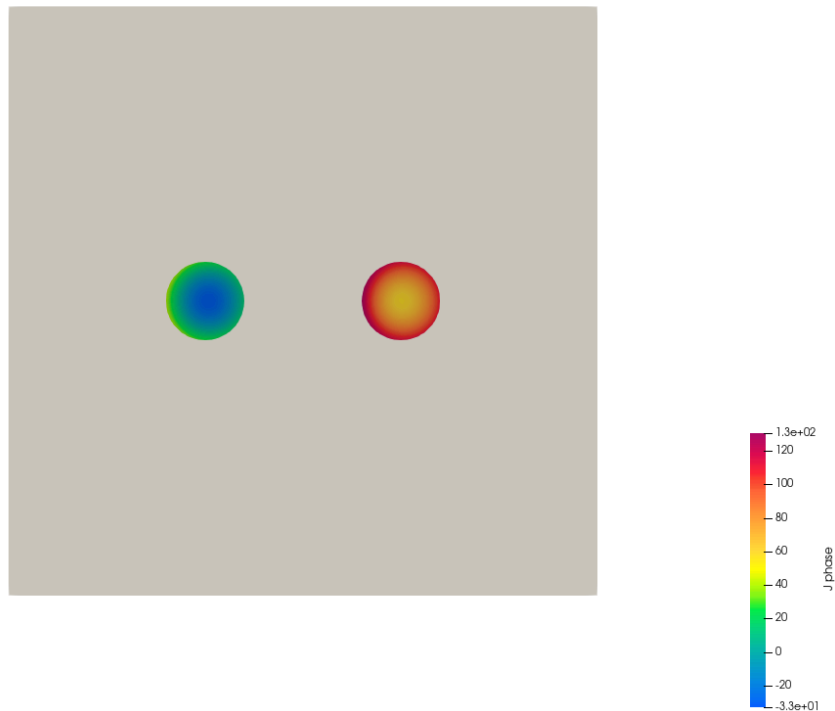
PP5 — the same, **phase_deg = 90**. A quarter cycle later
 the bright region has moved to the other wire. PP4 and
 PP5 are what **phase_deg** is.

The three panels are not on a common colour scale, and the colour bars say so.

`complex_magnitude` never reaches zero, so its dynamic range stays narrow and the renderer keeps the scale linear. `magnitude_at_phase` passes *through* zero wherever the instantaneous field nulls, so its range blows up and the renderer switches to logarithmic. The *structure* across the three is real; the relative brightness between them is not comparable. A pinned `range` key would fix this and the input format does not have one yet.

4.3 Phase

PP8: field J, plane
phase, degrees
phase-independent, f = 2500 Hz



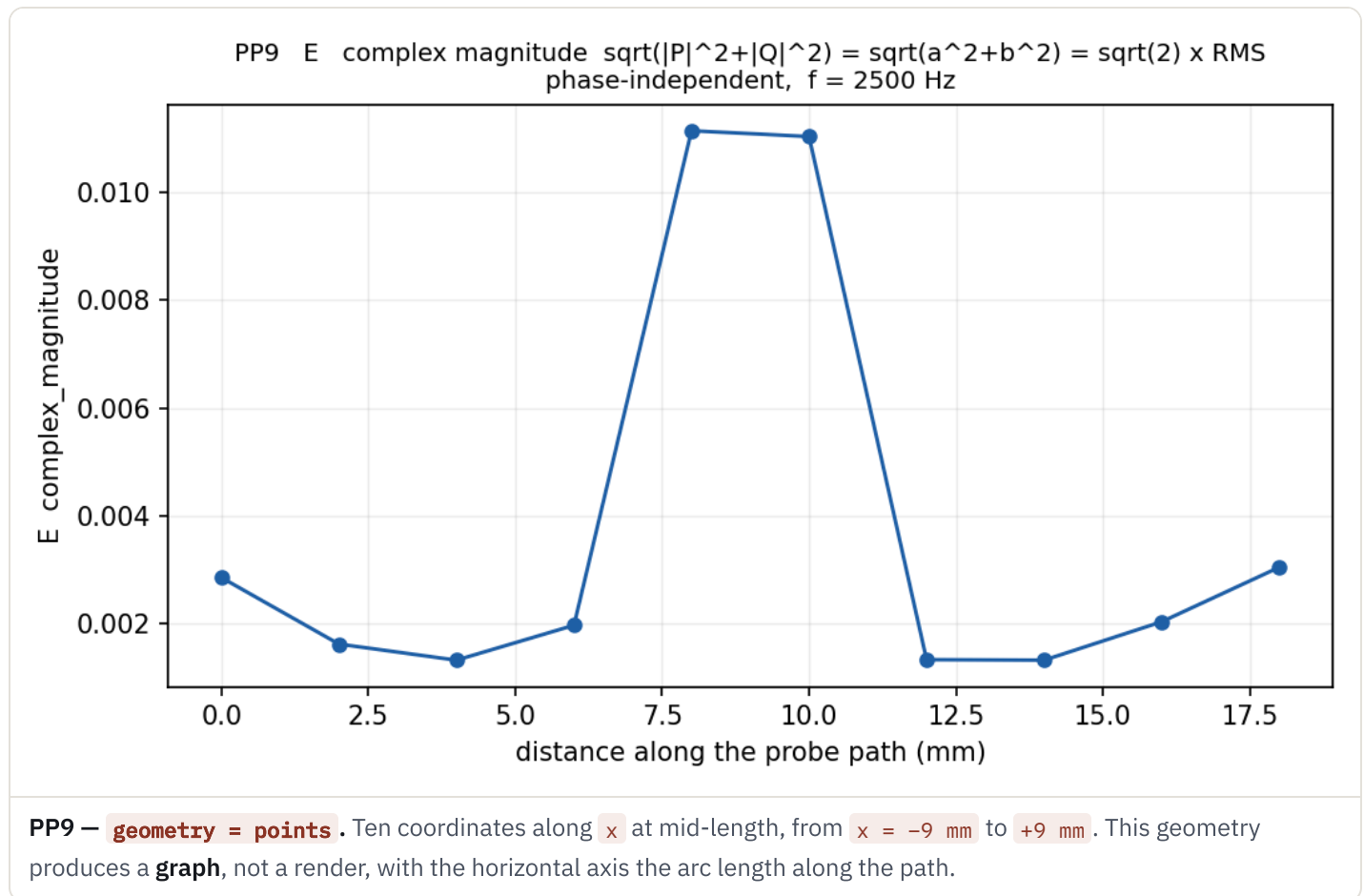
PP8 — `display = phase`, `component = z`. A vector field needs a component before a phase means anything, and the parser rejects `display = phase` on a vector without one. The two wires sit about 90° apart, and *within* each the phase varies with radius — the skin effect again, this time in the argument rather than the magnitude.

The air is blank on purpose. In an insulator `J` is identically zero, and a zero vector **has no phase** — it points nowhere. `atan2` returns an angle regardless, and *which* angle depends on the sign bit of the zero: `atan2(+0.0, -0.0) =  $\pi$` , and 21,727 of this mesh's 37,706 air nodes hold *negative* zero. The first version of this figure painted half the frame a confident, uniform **+180°**.

The renderer now writes NaN wherever the magnitude is negligible relative to the field's own scale, and sets the NaN colour to **white** — deliberately a colour the rainbow scale cannot produce, so a reader cannot mistake it for a reading. It arrives on the page as the flat neutral **grey** you see, RGB (200, 197, 189) against the figure's white margin, because ParaView shades the slice like any other lit surface. Either way it is outside the scale.

The solver's output contains no NaN or Inf anywhere — that was checked directly. This is the renderer declining to invent a phase for a quantity that does not have one.

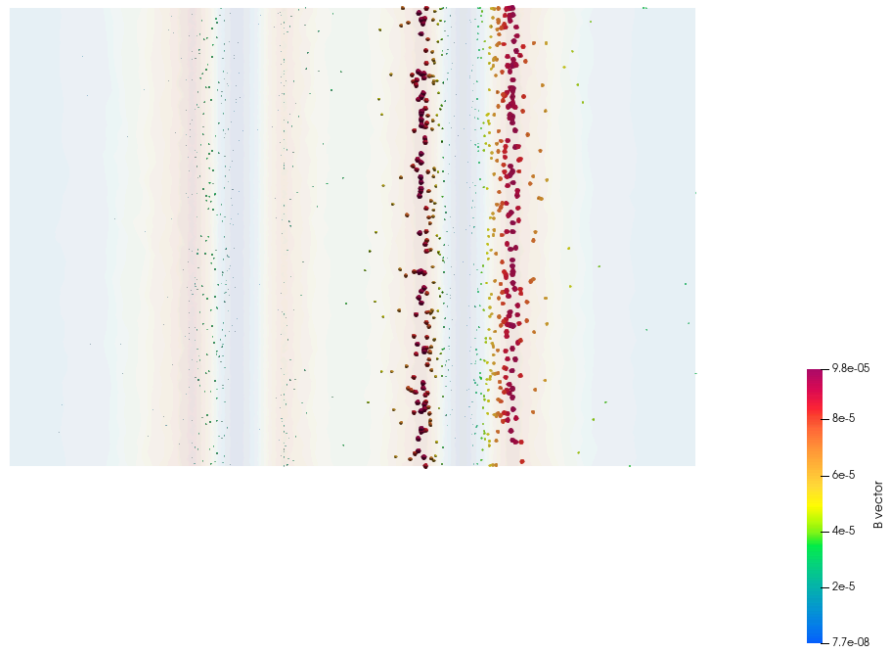
4.4 A line plot, and the other geometries



Inside each conductor $E \approx 1.3\text{e-}3 \text{ V/m}$, which is just J/σ : 1 A spread over $\pi a^2 = 12.6 \text{ mm}^2$ is $7.96\text{e}4 \text{ A/m}^2$, and $7.96\text{e}4 / 5.8\text{e}7 = 1.37\text{e-}3$. The **peak is in the air gap between the wires**, about $1.1\text{e-}2 \text{ V/m}$ — eight times the value inside the metal. That is the induced part, $E = -j\omega A - \nabla\Phi$: in the gap both wires' vector potentials contribute, and A falls away toward the box walls where $n \times A = 0$ pins it, which is why the two end points are lower again.

Four of these ten points sit exactly on a conductor surface — $x = \pm 3$ and ± 7 mm are the wire boundaries. The probe interpolates **nodal** data, and the solver's own output header flags those nodes `iface = 1` and says E is **meaningless** there, because its normal component genuinely jumps across a material interface. The figure is the renderer faithfully reporting what it was asked for; the four interface values should not be read as field values. Move the probe points off the surfaces, or read only the six clear of them.

PP10 field B plane
instantaneous vector P cos(th) - Q sin(th)
at phase wrt = 0 deg, f = 2500 Hz



PP10 — `plane = zx`, with no `offset`, so the cut is $y = 0$, the plane through both wire axes. The glyphs render as dots, and that is correct. This figure needs decoding, so the text below accounts for everything in the frame.

Three things are drawn, and only one of them is the data. The pale rectangle is the slice itself, shown as a surface at **12 % opacity** purely as a backdrop — the washed-out vertical stripes are $|B|$ on the cut, bleached nearly white. The dots are **arrow glyphs**:

`Glyph(GlyphType = "Arrow")` on every ~25th mesh point (`Stride = n // 1500`, so about 1,500 of the 38,579 points on this slice), each oriented along B and scaled *and* coloured by $|B|$. The colour bar is the third thing, and it reads correctly.

They look like dots because they point at you. On $y = 0$, B is almost perfectly normal to the plane:

MEAN $|B|$ OVER THE PLANE

$ B_x $	4.15e-7 T
$ B_y $	3.57e-5 T
$ B_z $	1.27e-7 T

Median $|B_y|/|B| = 0.9999$, and **94.3 % of the points have $|B_y|/|B| > 0.99$** . B around a z-directed current is azimuthal, and the plane through both axes is exactly the surface on which azimuthal means *straight out of the page*. Every arrow is therefore seen end-on — you are looking down the barrel at the cone head — and a bigger, redder dot simply means a larger $|B|$, because the Arrow glyph scales uniformly in all three dimensions.

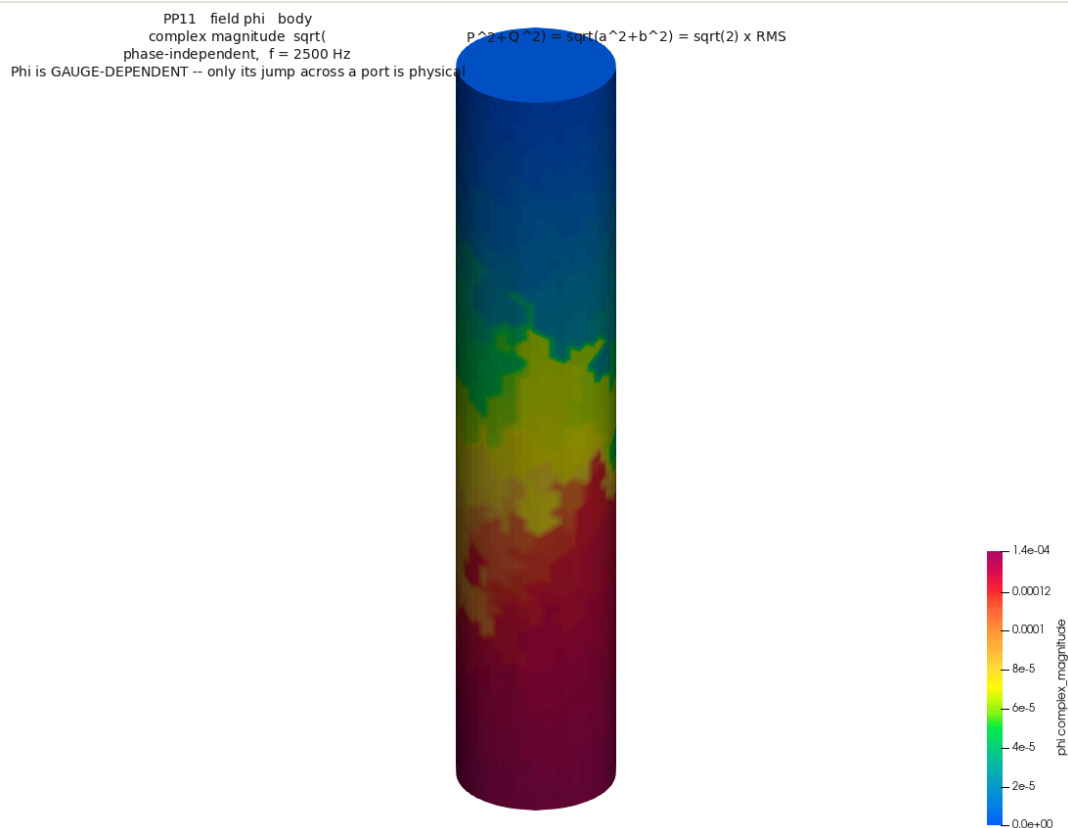
It is not a short-arrow artifact, which was checked separately: the median $|B|$ on the plane is 26 % of the maximum and the 75th percentile is 57 %, so a typical arrow has real length. The degeneracy is in orientation, not scale.

What the dots say once decoded. The camera sits at $+y$ looking along $-y$ with $+z$ up, so screen-right is $\hat{f} \times \hat{u} = (0, -1, 0) \times (0, 0, 1) = (-1, 0, 0)$: **$+x$ runs leftward**, putting wire 1 ($x = -5$ mm) on the *right* of the image and wire 2 ($x = +5$) on the left. Measured across the cut:

X BAND (MM)	MEAN $ B $ / MAX	
-9 ... -7	0.841	wire 1's outer surface
-7 ... -5	0.584	inside wire 1
-5 ... -3	0.401	inside wire 1
-3 ... -1	0.827	wire 1's inner surface
+3 ... +5	0.066	inside wire 2
+5 ... +7	0.254	inside wire 2

The **two dense dark-red columns are wire 1's two surfaces cut edge-on** — the same $|B|$ peak that appears as a red ring in PP3, sliced through rather than viewed down the axis — and $|B|$ sags between them because it falls toward the wire's own axis. The dim left half is **wire 2 at its zero crossing**: mean $|B|$ / max is 0.166 there against 0.552 on wire 1's half, the same quadrature asymmetry as PP4.

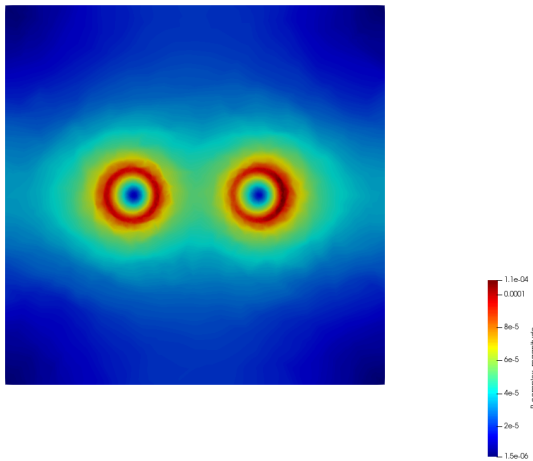
The useful negative result of the tour. `display = vector` is readable only when the plane is chosen to *contain* the field. The data here is right and only the glyph direction is degenerate. For this geometry the `xy` plane of PP3–PP6 is the one that shows `B`; `zx` is where `J` would be worth drawing instead.



PP11 — `field = phi`, a scalar, so no `component` is needed. Note the caption the renderer stamps on it: Φ is gauge-dependent. Its value depends on which spanning tree the tree-cotree gauge happened to pick, not only on the problem — a different tree shifts Φ by $-j\omega\psi$. `E`, `B`, `H` and `J` are the quantities that may be quoted; Φ and `A` are bookkeeping. §7 is the evidence for that, including why this case's $\omega L/R = 4.48$ puts PP11 firmly in the regime where Φ is mostly gauge.

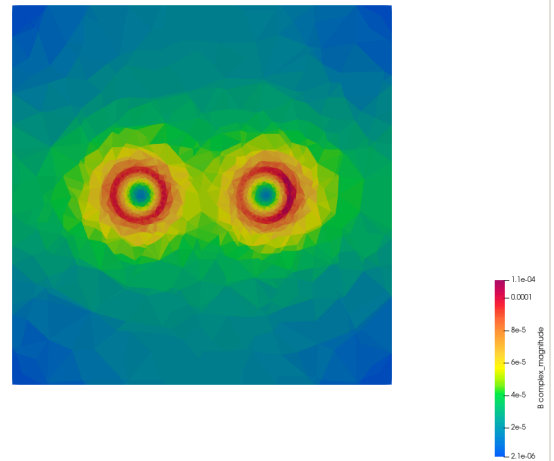
4.5 The presentation options

PP12 field B plane
complex magnitude sqrt($P^2 + Q^2 = \text{sqrt}(a^2 + b^2) = \text{sqrt}(2) \times \text{RMS}$)
phase-independent, $f = 2500$ Hz



PP12 — PP3 again with `colormap = jet`. Nine maps are available; `rainbow` (Rainbow Uniform) is the default, chosen because it reads as a *scale* at a glance without either end disappearing into the background.

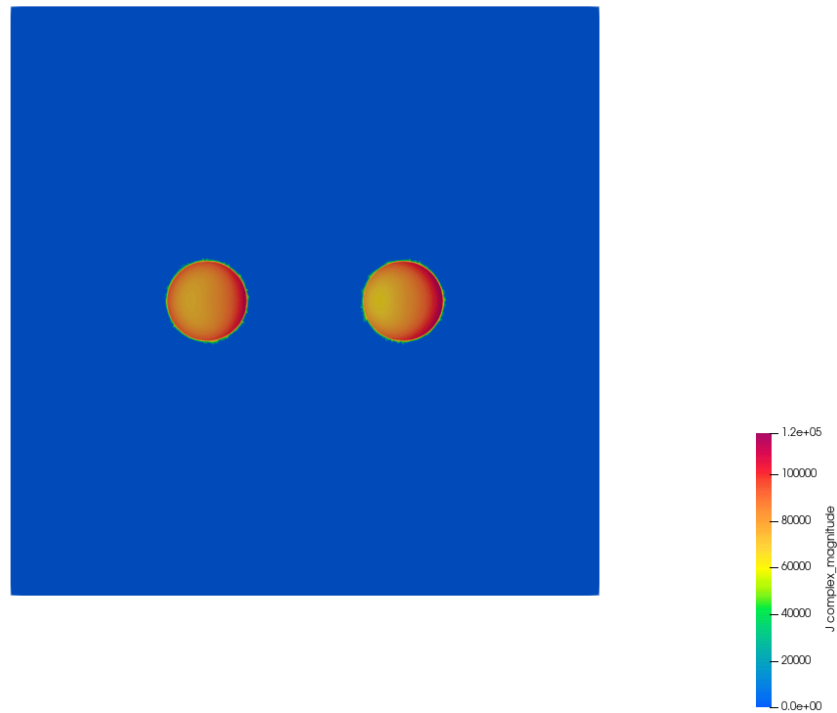
PP13 field B plane
complex magnitude sqrt($P^2 + Q^2 = \text{sqrt}(a^2 + b^2) = \text{sqrt}(2) \times \text{RMS}$)
phase-independent, $f = 2500$ Hz



PP13 — PP3 again with `data = per_tet`. One flat fill per element against PP3's smooth interpolation. Per-tet is not simply the uglier option: with first-order Whitney edge elements **B is exactly constant per tetrahedron**, so the cell array *is* the computed answer and the nodal one is an average of it. Nodal is the default because it reads better; per-tet is the honest one near a material interface, where the nodal average blends two values the field genuinely jumps between.

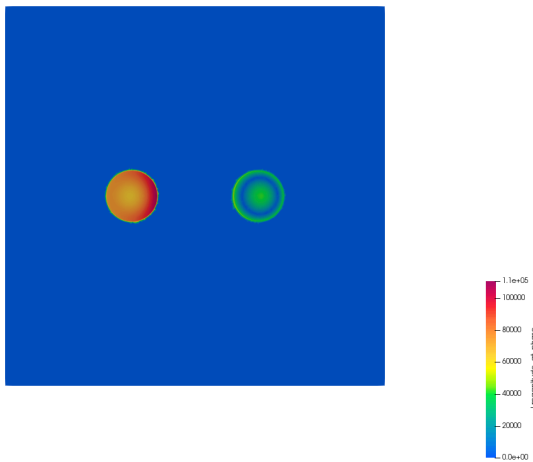
4.6 The same comparison, for the current

PP14 field J plane
complex magnitude $\sqrt{P^2 + Q^2} = \sqrt{a^2 + b^2} = \sqrt{2} \times \text{RMS}$
phase-independent, $f = 2500 \text{ Hz}$



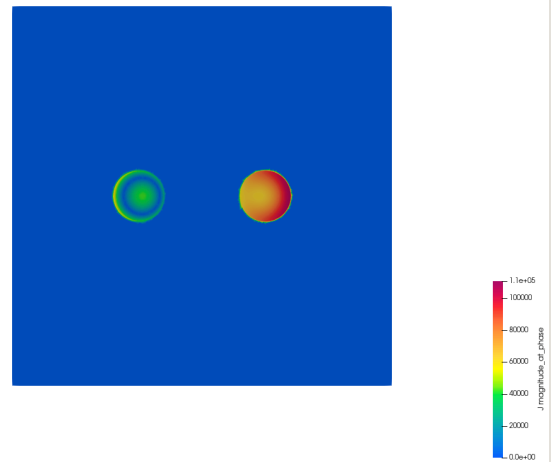
PP14 — **J**, phase-independent. Symmetric, both wires equally bright.

PP15 field J plane
instantaneous magnitude $P \cos(\theta) - Q \sin(\theta)$
at phase wt = 0 deg, $f = 2500 \text{ Hz}$



PP15 — **phase_deg = 0**. Wire 1 carries its full current while wire 2 sits at its zero crossing.

PP16 field J plane
instantaneous magnitude $P \cos(\theta) - Q \sin(\theta)$
at phase wt = 90 deg, $f = 2500 \text{ Hz}$

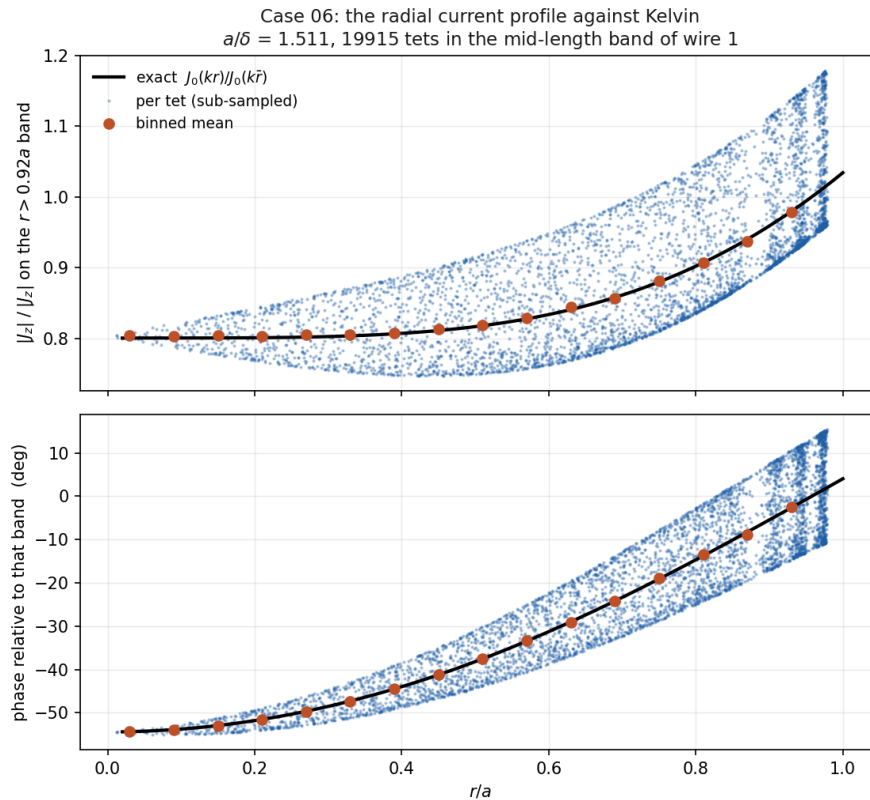


PP16 — **phase_deg = 90**. A quarter cycle later they swap.

The current is the quantity being *driven*, so PP15 and PP16 are the most direct view of what a port **phase_deg** does. PP4 and PP5 show the same thing one step removed, in the field those currents produce.

5. Skin-depth validation

5.1 The radial profile against Kelvin



The radial profile of J_z , magnitude above and phase below. Faint blue dots are individual tetrahedra in the mid-length band of wire 1 (sub-sampled); orange circles are the mean within each radial bin; the black curve is Kelvin. The azimuthal spread of the blue cloud at any radius is the proximity effect of §5.2.

Inside a straight round conductor the diffusion equation $\nabla^2 J_z = j\omega\mu\sigma J_z$ reduces to Bessel's equation, whose bounded solution is

$$\frac{J_z(r)}{J_z(a)} = \frac{J_0(kr)}{J_0(ka)}, \quad k = \frac{1-j}{\delta}, \quad \delta^2 = \frac{2}{\omega\mu\sigma}$$

At 2.5 kHz in copper $\delta = 1.3217 \text{ mm}$ and $a/\delta = 1.511$. Measured against that, per `tools/skin_profile.py`:

R/A	MEASURED	KELVIN	ERR	LAG	KELVIN	ERR
0.00–0.20	0.8034	0.8006	+0.35 %	–53.09°	–53.05°	–0.04°
0.20–0.40	0.8048	0.8028	+0.24 %	–47.86°	–47.86°	+0.00°
0.40–0.60	0.8182	0.8171	+0.14 %	–37.54°	–37.56°	+0.03°
0.60–0.80	0.8624	0.8621	+0.04 %	–22.64°	–22.71°	+0.07°
0.80–0.92	0.9343	0.9349	–0.06 %	–8.78°	–8.91°	+0.13°

A 53° lag from surface to core, reproduced to 0.04°, and the magnitude profile to better than 0.35 % everywhere. Both sides are anchored on the same $r > 0.92a$ band, whose mean radius is 0.9575a — §5.3 is about why that sentence matters more than it looks.

tools/skin_profile.py reduces any case's output the same way, which makes the comparison portable:

	A/Δ	ELEMENTS PER Δ	CORE J ERR	WORST LAG ERR
04, single cylinder	2.998	3.2	+1.20 %	0.39°
06, this case	1.511	4.2	+0.35 %	0.13°

Case 06 is the **better-resolved** of the two per skin depth; case 04 is the **harder problem** — 138° of rotation from surface to core against 53°, and |J| down to 27 % of its surface value against 80 %.

5.2 Proximity effect, which Kelvin does not contain

	J FACING THE NEIGHBOUR / AWAY FROM IT
wire 1	1.137
wire 2	0.766

The neighbour's field redistributes current around the circumference by +14 % and –23 %. It is what makes PP1's arrows asymmetric. The two wires are pushed **opposite** ways precisely because they are 90° apart: at any instant one is near its peak while the other is at its zero crossing, so what each sees from the other is not the same thing.

Kelvin's solution is axisymmetric and contains none of this, so the comparison in §5.1 uses the **azimuthal mean** at each radius, which removes it to first order. The proximity effect is real physics the case captures and the analytic reference does not describe — it is not an

error, and it is not something the §5.1 table is testing.

5.3 Both sides must be reduced identically

This is the most useful thing the case taught, and it cost two wrong results.

Kelvin's profile is normalised at $r = a$. A measurement cannot be: a cell band "at the surface" is $r > 0.92a$, whose **mean radius is $0.9575a$** , because cell centres never reach the boundary. Dividing the measurement by $J(0.9575a)$ while dividing the theory by $J(a)$ compares two profiles pinned at different places, and inflates every measured ratio by about **+3.5 %**.

That offset contains no h , so no amount of refinement can remove it — and it told a thoroughly convincing story: core errors of 7.2 % that "improved" to 3.9 % under refinement and then stalled, an apparent convergence order collapsing to 0.37, and a tidy physical explanation for the floor (a modelling error from the proximity effect, or from the finite length). All of it was one constant offset sitting on a real error ten times smaller. The retracted 7 % figure was quoted in `two_wires.geo`'s own header until this report was written.

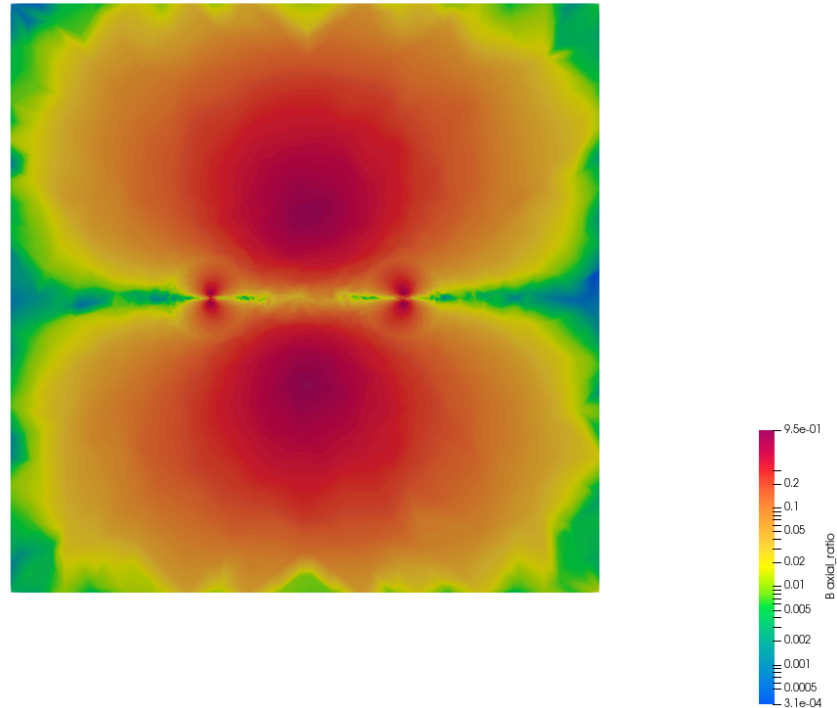
Two controls placed it. A **single isolated wire** — same radius, length, box and element size, no neighbour — showed the *identical* error, so not the proximity effect. Narrowing the axial sample band from 0.30–0.70 `Lz` to 0.45–0.55 `Lz` changed the fourth decimal, so not the ends. **An error that survives removing the neighbour, ignores axial position, and ignores element size is in the instrument, not the solve.**

A second, subtler version of the same mistake followed. The measured reference is a *mean over cells*, so the Bessel reference must be the **mean of J_0 over those same radii** — not J_0 evaluated at their mean radius. J_0 is not linear, so the two differ by its curvature across the band. Case 04's own `make_plots.py` had always done this correctly; the shared tool had not.

A third, found while writing §7: the radius itself. Inferring a from a percentile of cell-centre radii under-reads it by about 2 %, because the outermost centre sits half an element inside the surface — and under-reading a inflates the reported a/δ . The cross-section **area** is unbiased: $a = \sqrt{(\text{area}/\pi)}$, which is what both scripts now use, and `tools/skin_profile.py` prints the ratio of the two as a roundness check.

6. The polarization ellipse

The payoff, and the reason for the geometry.



PP7 — `display = axial_ratio` : the *shape* of the polarization ellipse, `b/a` , which is 0 for a linearly polarized field and 1 for a circular one. Two lobes above and below the line of centres, and a null along it.

Both features are what the two-wire superposition predicts. A phasor vector traces `B(u) = P cos u - Q sin u` ; with `I1 = I` and `I2 = jI` the real part `P` is the field of wire 1 alone and the imaginary part `Q` is the field of wire 2 alone. So:

`b/a = 0` on the line of centres – the two wires' fields are PARALLEL there, and a sum of two parallel vectors out of phase is still linear

`b/a = 1` at $(0, \pm d/2)$ – the two contributions are perpendicular and equal, which is a circle

	MEASURED	EXACT
max <code>b/a</code> near $(0, \pm d/2)$	0.9647	1
<code>N/a</code> at the most circular cell	1.3895	<code>√2 = 1.41421</code>
median <code>b/a</code> – analytic, per cell, near the axis	-0.0098	0

`N/a = √2` is the equality case of the chain `b ≤ |B(θ)| ≤ a ≤ N ≤ √2 a` , derived in `docs/ComplexVectorPhasorConcept.md` . Where the field is circularly polarized, `complex_magnitude` exceeds the true peak by **41 %** — so PP3 and PP6 differ by that much

at those two points and not at all on the line of centres. That is the whole of the phasor document in two pictures.

Read **axial_ratio** next to a magnitude plot, never alone. Here the ellipticity sits where the field is strong: the median **b/a** is 0.140 among cells carrying at least 30 % of peak **|B|**. In case 05 it was the opposite — the most circular cell there carried **8.1e-04** of the peak, three decades down and physically invisible. A nearly circular but negligible field is indistinguishable from a nearly circular dominant one on this plot, because the plot shows only shape.

7. Φ , the gauge, and what "voltage" means here

PP11 carries a warning stamped by the renderer — **Φ is gauge-dependent**. This section is what stands behind it, because the claim is sharper than it sounds and the obvious reading of it is wrong. Everything here was measured on **01_OneCylinder**, whose small mesh makes a three-way comparison cheap, and then anchored back to this case.

7.1 The freedom, and what survives it

A and Φ are not unique. For any scalar field **ψ** ,

$$A \rightarrow A + \nabla\psi \qquad \Phi \rightarrow \Phi - j\omega\psi$$

leaves every field unchanged. **B** survives because the curl of a gradient is identically zero:

$$B = \nabla \times A \rightarrow \nabla \times (A + \nabla\psi) = \nabla \times A + \nabla \times \nabla\psi = \nabla \times A = B$$

and **E** survives because the two shifts cancel against each other:

$$\begin{aligned} E = -j\omega A - \nabla\Phi &\rightarrow -j\omega(A + \nabla\psi) - \nabla(\Phi - j\omega\psi) \\ &= -j\omega A - j\omega\nabla\psi - \nabla\Phi + j\omega\nabla\psi \\ &= -j\omega A - \nabla\Phi = E \end{aligned}$$

$H = B/\mu$ and **$J = \sigma E$** inherit it. **Neither potential is invariant on its own**; only the combination that forms **E** is, which is why A and Φ must move together.

This is exact in the discrete system, not only in the continuum, and that is precisely what licenses tree-cotree gauging. The gradient of a nodal function is *exactly* representable in the Whitney edge space — on edge (i,j) its degree of freedom is just $\psi_j - \psi_i$ — so the discrete gauge freedom is exactly the set of discrete gradients and the discrete $\nabla \times \nabla \psi$ is exactly zero. Nothing in the cancellation is a discretisation approximation.

7.2 Measured, on three spanning trees

The tree-cotree gauge picks one representative by forcing $A = 0$ on a spanning tree's edges. A **different tree picks a different representative**, so solving the same problem on two trees is a direct test.

Different trees were obtained by permuting the order the nodes are listed in the `.msh`. The reader assigns each node its internal index in file order, edges are numbered by sorted vertex pair, and the tree is rooted at the lowest-numbered Dirichlet node — so reordering the file changes the tree while the mesh stays geometrically identical. That it worked is confirmed by the solver: identical 23,547 tets and 55,652 unknowns, but **1,144,897 / 1,145,056 / 1,145,083 stored nonzeros** — different edges gauged away.

	MAX VALUE	MAX CHANGE	RELATIVE
Φ potential	1.0000e+00	1.5186e-01	1.52e-01
A potential	1.8376e+00	3.9800e-01	2.17e-01
B field	2.8329e+00	4.5522e-12	1.61e-12
E field	3.7594e+01	6.1495e-12	1.64e-13
H field	2.2544e+06	3.6226e-06	1.61e-12
J field	4.9177e+08	3.0737e-04	6.25e-13

Both potentials move by 15–22 %. Every field is unchanged at 1e-12 to 1e-13 — eleven orders of magnitude between the two groups. The residual is not leftover gauge sensitivity: the three runs factorize genuinely different matrices, so the arithmetic happens in a different order, and 1e-13 is what double precision accumulates through a 1.1 M-nonzero factorization.

7.3 $\omega L/R$ decides how much of Φ is gauge

$\omega L/R$ does not make Φ gauge-dependent — Φ is gauge-dependent at every frequency.

What it sets is **the size of the gauge term relative to the part you wanted to read.**

Integrate the field definition along the conductor:

$$\nabla\Phi = -E - j\omega A$$

$$\begin{aligned} \Phi(L) - \Phi(0) &= -\int E \cdot dl - j\omega \int A \cdot dl \\ &\quad \underbrace{\quad\quad\quad}_{\approx I \cdot R} \quad \underbrace{\quad\quad\quad}_{\approx j\omega L \cdot I} \\ &\quad \text{gauge-INVARIANT} \quad \text{gauge-SENSITIVE} \end{aligned}$$

The first term is built from E , invariant to $1e-13$ by the table above. The second is built from A , which is exactly what the gauge shifts — under $A \rightarrow A + \nabla\psi$ it changes by $j\omega[\psi(\text{end}) - \psi(\text{start})]$. Their ratio is $\omega L/R$.

It **saturates**, because a contaminated fraction cannot exceed 1. The correct form is

$$\text{gauge-sensitive fraction} \approx \omega L/|Z| = \omega L/\sqrt{R^2 + (\omega L)^2} = \sin(\arg Z)$$

which was checked rather than assumed. Case 01 has $R = 2.26e-6 \Omega$, $L = 20.9 \text{ nH}$:

	$\omega L/R$	$\text{SIN}(\text{ARG } Z)$
50 Hz	2.89	0.945
1 Hz	0.0578	0.0577

predicted ratio **16.4**, against measured **15.9** (Φ vs z/L) and **16.0** (tree-to-tree shift) — agreement to 3 %. Note this is emphatically *not* 50: the plain $\omega L/R$ ratio would predict 50 and the measurement says 16, so the saturating form is the right one. The leftover factor of ~1.8 in the absolute value is the $0(1)$ geometric constant in $\psi \sim |A| \cdot \ell$, with ℓ the tree path length — which is tree-dependent, and that is the point.

Where each case sits:

CASE	$\Omega L/R$	Φ VS Z/L	Φ CHECKED?
02, 03	$\ll 1$	$8.26\text{e-}04 / 4.3\text{e-}06$	asserted
01 @ 50 Hz	2.89	0.488	deliberately omitted
06, this case	4.48	—	not applicable
04	12.8	0.51	deliberately omitted

This case's figure is from $R = 3.086\text{e-}5 \Omega$ taken from ohmic dissipation ($R = 2P/|I|^2$, giving $R/R_{dc} = 1.1245$ at $a/\delta = 1.511$) and $L = 8.80 \text{ nH}$. **R must come from dissipation here and not from V/I** : with $I_2 = jI_1$ the mutual term $j\omega M \cdot I_2 = -\omega M \cdot I_1$ is *real*, so it adds to one wire's driving-point resistance and subtracts from the other's — measured, $6.65\text{e-}5$ and $4.63\text{e-}6$, which straddle $R_{dc} = 2.74\text{e-}5$ and even put wire 1 *below* it. Dissipation gives $3.0857\text{e-}5$ and $3.0904\text{e-}5$ for the two wires, agreeing to 0.15 % as they must. The mutual term lands entirely outside L , which is why both wires report the same 8.80 nH.

So PP11 is firmly in the gauge-dominated regime, and reading its colour bar as a voltage would be a mistake.

Why a magnitude probe can still look right. At low ω the current is nearly in phase with the drive, so A is nearly real, so ψ is nearly real — and $-j\omega\psi$ is then nearly *purely imaginary*. It lands in $\text{Im}(\Phi)$ and leaves $\text{Re}(\Phi)$ almost clean. Measured inside the wire:

FREQ	MAX $ \Delta \text{RE}(\Phi) $	MAX $ \Delta \text{IM}(\Phi) $
1 Hz	$5.64\text{e-}04$	$9.50\text{e-}03$
50 Hz	$1.44\text{e-}01$	$4.96\text{e-}02$

At 1 Hz the contamination is 17× larger in the imaginary part; by 50 Hz it has flooded into the real part and dominates there. That is why $\text{Re}(\Phi)$ tracks Z/L to $1.9\text{e-}03$ at 1 Hz and to only $4.88\text{e-}01$ at 50 Hz — a factor of 254 on a 50× change in ω , steeper than the complex quantity's 16× because the real part only picks up a higher-order remainder.

7.4 The terminals are exact anyway — with a voltage *or* a current source

This is the part that matters in practice, and it refuted the natural guess.

The gauge function is not free everywhere. $\mathbf{n} \times \mathbf{A} = \mathbf{0}$ on the outer boundary forces ψ to be **constant** on it; a port that pins Φ forces $\psi = 0$ there. In this case and in case 01 the wires run the full height of the box, so **both terminals lie in the box's end faces**, and the box surface is connected — so $\psi = 0$ on the whole outer boundary, terminals included.

Recovering ψ from the measured shift ($\delta\psi = j\delta\Phi/\omega$) confirms it:

REGION	NODES	MAX $ \Delta\psi $ (WB)
the whole outer boundary	4,770	3.93e-21
of which, the driven terminal	477	3.83e-21
of which, the 0 V reference	477	0.00e+00
the interior	29,478	3.35e-09

Twelve orders of magnitude. **The gauge lives in the interior and vanishes on the boundary**, so it distorts the *shape* of Φ while pinning its endpoints.

With a current source this still holds, which is not obvious: a current-driven port pins Φ at the reference terminal only, leaving the driven terminal's potential a free unknown. Driving case 01's geometry with 1 A at $\omega L/R = 2.89$ — the gauge-dominated regime — on the same three trees:

TREE	$ V $ (V)	R (Ω)	L (H)
base	6.926620e-06	2.264520e-06	2.083654e-08
permA	6.926620e-06	2.264520e-06	2.083654e-08
permB	6.926620e-06	2.264520e-06	2.083654e-08

Identical. The change in V across trees is 1.2e-18 V — 0.00 % — and the driven face comes out an exact equipotential (spread 4e-22). The reason is 7.4's first paragraph: that terminal sits on the outer boundary, where ψ is pinned to zero regardless of which port type drives it. R and L also match the voltage-driven solve's 2.26e-6 Ω and 20.87 nH.

The caveat is the geometry, not the port type. This argument needs the terminal to lie on the outer boundary where $\mathbf{n} \times \mathbf{A} = \mathbf{0}$. An electrode floating in the *interior* of the domain would have no such protection, and its potential would be gauge-dependent. Every case in this suite puts its terminals on the boundary.

7.5 What to read, and what not to

Always safe

E , B , H , J ; terminal voltages; $\int E \cdot dl$ on a stated path; I , R , L , Z

Safe only when $\omega L/R \ll 1$

$\text{Re}(\Phi)$ as a z/L gradient inside a conductor

Never

Φ or A at an interior node, read as a physical potential

And one point that is physics rather than numerics: in a time-varying field the voltage between two points is genuinely **path-dependent** — that is Faraday's law. Any tool that reports a single "the potential here" has quietly chosen a path, or a gauge, on your behalf.

8. What is asserted automatically

`..\check.bat 06_TwoWires_Quadrature` runs `verify_two_wires.py`. Eight checks, all **passing**:

	MEASURED	TARGET
$ I $ in wire 1	1.000405	1 A $\pm$ 2 %
$ I $ in wire 2	0.998795	1 A $\pm$ 2 %
phase difference	90.014°	90° $\pm$ 1°
max b/a near $(0, \pm d/2)$	0.9647	≥ 0.90
N/a at the most circular cell	1.3895	$\sqrt{2} \pm 0.05$
median b/a – analytic, per cell	−0.0098	0 $\pm$ 0.03
core $ J $ vs Kelvin	0.80301	0.80028 $\pm$ 1 %
core phase lag vs Kelvin	−53.126°	−53.230° $\pm$ 0.5°

The b/a residual is compared **cell by cell**, with the analytic value evaluated at each cell's own centroid. An earlier version compared the cells' *median* against the analytic's *area* average over the same region: that assertion moved from 0.0658 to 0.0509 when the mesh was refined while no physics changed at all. **A test that moves with the mesh and not with the field is measuring the mesh.**

Run summary

Tets	192,368
Unknowns	448,438 — A: 190,175 free edges, 31,733 gauged to zero, 6,582 Dirichlet; Φ : the rest
Stored nonzeros	9,659,864
<code>nnz(L)</code> after AMD	384,933,274
Backward error	5.33e-22
Timing	assemble 0.75 s, analyze 5.0 s, factorize 10.8 s, solve 0.23 s

Reproducing

```
gmsh two_wires.geo -3 -o two_wires.msh
..\run_case.bat two_wires.aphi
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" ..\..\tools\postprocess.py output
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" make_report_figs.py output
"C:\Program Files\ParaView 6.1.1\bin\pvpython.exe" collect_postprocess_figs.py
sh build_doc.sh
```

`make_convergence_data.py` rebuilds §2.2's three meshes and `convergence.json`; it takes a few minutes and is only needed when the mesh changes. `tools/skin_profile.py <output-dir>` prints §5.1 for this or any other case.

Run `build_doc.sh` after regenerating figures, or `TwoWiresValidation.html` and `.pdf` keep showing the previous run's pictures — they are inlined, not referenced.